

SHIVAM SINGH

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EDUCATION

Indian Institute of Information Technology, Design and Manufacturing Kurnool

(Aug 2023 - June 2027)

B.Tech, Artificial Intelligence and Data Science; CGPA: 8.11/10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Design, Operating Systems, Database Management Systems, Distributed Systems, Complexity Analysis, Linear Algebra & Optimization

TECHNICAL SKILLS

Languages: Java, Python, C, SQL, JavaScript, HTML/CSS

CS Fundamentals: DSA, OOD, System Design, Database Design, Operating Systems, Distributed Systems

Web/Backend & AI: React, Redux, Node.js, Express, FastAPI, MongoDB, Docker, LLM APIs, RAG, Vector Databases, LangChain

Cloud & Tools: AWS, GCP, Git, VS Code, Postman, Agile/Scrum

EXPERIENCE

Caterworld.ai — Software Development Intern, Remote

(Apr 2026 - Present)

- Accelerated caterer onboarding completion by 15-20%, measured by user activation, by redesigning onboarding UI for this commercial product in React.js, TypeScript, and PostgreSQL within an Agile team.
- Eliminated unauthorized-access incidents post-launch by engineering a hierarchy-based multi-user approval system with role-based access control, applying systems analysis and design to model permission structures.
- Enabled location-based routing across Telangana, Karnataka, and Andhra Pradesh by architecting a pin-code mapping system backed by relational database design best practices.
- Improved sprint delivery predictability, measured by on-time feature ship dates, by collaborating in Agile ceremonies to deliver features in a production software environment.

PROJECTS

AI-Powered Hotel Booking & Concierge Platform | MERN, Python, FastAPI, LLM, RAG

(May 2025 - Present)

Live Demo: <https://rr-palace2.vercel.app/>

- Architected an end-to-end hotel booking platform using MERN, designed to support 500+ monthly users, with JWT/Bcrypt authentication and RBAC-based admin controls for room inventory, reservations, and operational analytics.
- Integrated a Python/FastAPI AI service with LLM and RAG, using embeddings and vector search over hotel policies, room information, and FAQs to provide context-aware responses and personalized room recommendations.
- Implemented multi-turn conversational memory and tool calling to connect the AI assistant with existing hotel APIs for room discovery, availability, pricing, and customer queries, while keeping core business logic and validation within the backend.
- Engineered reliable booking workflows with concurrency controls and automated background jobs to prevent race conditions during simultaneous bookings and release unconfirmed "ghost" reservations, achieving a 98/100 Lighthouse SEO score in low-bandwidth mobile environments.

Spatio-Temporal Federated Meta-Learning for EV Occupancy Prediction | React, Node.js, PyTorch

(Feb 2026 - Present)

- Architected a distributed, fault-tolerant big data ML framework across edge nodes for EV occupancy prediction, deployed on GCP cloud platform; cut end-to-end latency 40% via Asynchronous Global Update algorithms.
- Engineered a custom aggregation server implementing Adaptive Reptile training and staleness-aware weighted aggregation, enabling continuous global model updates without blocking on slow edge clients.
- Reduced end-to-end aggregation latency by decoupling edge training from central aggregation, utilizing configurable batch buffering and tuning Kafka payload buffers for robust ~1.5MB weight transfers.
- Built a central command dashboard (React/Node.js) monitoring real-time node health and weight transmissions across distributed nodes.

ACHIEVEMENTS

- 1st Place, DataWorks Club Hackathon, IIITDM Kurnool Fest (SOLASTA 2026) - built a data-driven prediction model.
- Solved 600+ LeetCode problems (1580 rating) and ranked under 900 in CodeChef Starters.
- State Winner, International Mathematics Olympiad (IMO).

POSITIONS OF RESPONSIBILITY

NSS Unit, IIITDM Kurnool — Branch Coordinator

(Jul 2024 - Present)

- Directed 10+ community events and surveys, coordinating logistics for 200+ active participants.

College Fest (SOLASTA) — Backstage Member

(Jan 2024 - Feb 2025)

- Coordinated stage operations and artist logistics for 1000+ attendees with zero downtime between shifts.